THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The property of metals by which they can be beaten into thin sheets is: 1

(a) brittleness **(b)** malleability **(c)** ductility **(d)** sonority

2. Which of these solutions is a solid in a solid? 1

(a) soda water **(b)** air **(c)** vinegar **(d)** brass

3. The reflection from a rough surface in which parallel rays are reflected in different directions is: 1

(a) total reflection **(b)** refraction

(c) regular reflection **(d)** diffuse reflection

4. The Moon takes about ___ to go around the Earth once. 1

(a) 27 days **(b)** 7 days **(c)** 365 days **(d)** 24 hours

5. A series of organisms in which each one is eaten by the next is a: 1

(a) habitat **(b)** food web **(c)** ecosystem **(d)** food chain

6. The most abundant gas in the Earth's atmosphere is: 1

(a) nitrogen **(b)** carbon dioxide **(c)** argon **(d)** oxygen

7. In an investigation of how the length of a pendulum affects its time period, the independent variable is: 1

(a) colour of the bob **(b)** time period

(c) mass of the room **(d)** length of the pendulum

8. Plant cells usually have a large central: 1

(a) vacuole **(b)** membrane only **(c)** chloroplast only **(d)** nucleus only

9. Scurvy is caused by deficiency of: 1

(a) iron **(b)** vitamin C **(c)** calcium **(d)** vitamin A

10. A zigzag line in a circuit diagram represents: 1

(a) a switch **(b)** a cell **(c)** a resistor **(d)** a bulb

11. A body is charged by rubbing. The two types of charges are: 1

(a) north and south **(b)** hot and cold

(c) positive and negative **(d)** heavy and light

12. A cyclone has at its centre a region of: 1

(a) no air **(b)** very low pressure
(c) solid ice **(d)** very high pressure

13. Which of the following increases the rate of evaporation? 1

(a) increase in temperature **(b)** increase in humidity
(c) decrease in surface area **(d)** absence of wind

14. A mixture is: 1

(a) a pure substance
(b) always a liquid
(c) a combination of two or more substances not chemically combined
(d) a single compound

15. The mass percentage of a solution is: 1

(a) mass of solute x mass of solvent
(b) (mass of solute / mass of solution) x 100
(c) (mass of solvent / mass of solute) x 100
(d) (mass of solution / mass of solute) x 100

16. The relation between the focal length (f) and the radius of curvature (R) is: 1

(a) R = f / 2 **(b)** f = 2R **(c)** R = 2f **(d)** R = f

17. The calendar based on the motion of the Earth around the Sun is the: 1

(a) lunar calendar **(b)** lunisolar calendar
(c) weekly calendar **(d)** solar calendar

18. The process of conversion of dead matter into simple substances by decomposers is: 1

(a) pollination **(b)** decomposition **(c)** respiration **(d)** photosynthesis

19. The Earth's water that is available as fresh water for use is: 1

(a) all the ice **(b)** all the oceans
(c) a small part of the total water **(d)** most of the water

20. In the same study, the time taken by the puri to puff up is the: 1

(a) control factor **(b)** hypothesis
(c) independent variable **(d)** dependent variable

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. State two assumptions of the particulate model of matter. 2

22. Define an element. Give two examples. 2

23. Define solute, solvent and solution. 2

24. State the laws of reflection. 2

25. Why does the Sun appear to rise in the east and set in the west? 2

26. What is an ecosystem? Give two examples. 2

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SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Describe an activity to show that the solubility of a solid increases with temperature. 3

28. Draw in words the ray diagram of reflection at a plane mirror and mark i and r. 3

29. Describe an activity to make a simple sundial and use it to measure time. 3

30. Draw in words a pond ecosystem and name its biotic and abiotic components. 3

31. Describe the layers of the Earth with one feature of each. 3

OR

Why is the scientific method not a rigid list of steps? Explain with a situation. 3

32. Why is the cell called the structural and functional unit of life? 3

OR

Name the deficiency diseases caused by lack of vitamin A, vitamin C and iron, giving one symptom of each. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. State and explain the laws of reflection with a labelled description. Verify the laws using an activity. 5

34. Describe how the Earth's rotation and revolution lead to day and night, years and seasons. 5

35. Describe the components of an ecosystem with examples and show how they are interdependent. 5

36. Describe the interior of the Earth with the layers, their states and composition, and how scientists know about the interior. 5

OR

Describe how errors can occur while taking measurements and how they can be reduced. 5

37. Differentiate between bacteria, viruses, fungi and protozoa in a tabular form (any four points). 5

OR

How do personal and community hygiene help in staying healthy? Give at least three points for each. 5

38. Explain the use of electromagnets in cranes and for separating iron. Why is soft iron used? 5

OR

Explain the working of a gold-leaf electroscope and what happens when a charged rod touches or is brought near the disc. 5

END OF QUESTION PAPER — Total: 80 Marks

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